

Section 12.5

1. 15 points Let $f(x) = x^2 - 2$. Find the absolute minimum and the absolute maximum of $f(x)$ over the interval $[-5, 5]$, if such an absolute maximum or minimum exists. For every absolute extremum, give the point $(x, f(x))$ and indicate whether it is a minimum or maximum.

First, we find $f'(x)$:

$$f'(x) = 2x.$$

Then, we set $f'(x) = 0$ and solve to find the root $x = 0$. Following this, we find $f''(x)$:

$$f''(x) = 2.$$

As such, $f''(0) = 2 > 0$, and f is concave upward at $x = 0$. Noting that 0 is indeed in $[-5, 5]$, we then conclude that $(0, f(0)) = (0, -2)$ is indeed an absolute minimum of f on the interval $[-5, 5]$. Checking the endpoints, we then reach that $(-5, f(-5)) = (-5, 23)$ and $(5, f(5)) = (5, 23)$ are absolute maximums of f on the interval $[-5, 5]$.

2. 15 points Let $f(x) = x^2 - 2$. Find the absolute minimum and the absolute maximum of $f(x)$ over the interval $[2, 3]$, if such an absolute maximum or minimum exists. For every absolute extremum, give the point $(x, f(x))$ and indicate whether it is a minimum or maximum.

First, we find $f'(x)$:

$$f'(x) = 2x.$$

Then, we set $f'(x) = 0$ and solve to find the root $x = 0$. Following this, we find $f''(x)$:

$$f''(x) = 2.$$

As such, $f''(0) = 2 > 0$, and f is concave upward at $x = 0$. Noting that 0 is not in $[2, 3]$, we disregard this point.

Checking the endpoints, we then reach that $(2, f(2)) = (2, 2)$ is an absolute minimum of f on the interval $[2, 3]$, and $(3, f(3)) = (3, 7)$ is an absolute maximum of f on the interval $[2, 3]$.

3. 15 points Let $f(x) = \ln(x) - \frac{1}{2}x$. Find the absolute minimum and the absolute maximum of $f(x)$ over the interval $(1, 20)$, if such an absolute maximum or minimum exists. For every absolute extremum, give the point $(x, f(x))$ and indicate whether it is a minimum or maximum.

First, we find $f'(x)$:

$$f'(x) = \frac{1}{x} - \frac{1}{2}.$$

Then, we set $f'(x) = 0$ and solve to find the root $x = 2$. Following this, we find $f''(x)$:

$$f''(x) = -\frac{1}{x^2}.$$

As such, $f''(2) = -\frac{1}{4} < 0$, and f is concave downward at $x = 2$. Noting that 2 is in the interval $(1, 20)$, we conclude that $(2, f(2)) = (2, \ln(2) - 1)$ is an absolute maximum of f on the interval $(1, 20)$.

As $(1, 20)$ is an open interval, there are no endpoints to check, so we conclude that $f(x)$ has no absolute minimum on $(1, 20)$.

4. 15 points Let $f(x) = x^3 - 3x^2 + 1$. Find the absolute minimum and the absolute maximum of $f(x)$ over the interval $[1, 15]$, if such an absolute maximum or minimum exists. For every absolute extremum, give the point $(x, f(x))$ and indicate whether it is a minimum or maximum.

First, we find $f'(x)$:

$$f'(x) = 3x^2 - 6x.$$

Then, we set $f'(x) = 0$ and solve to find the roots $x = 0$ and $x = 2$. Following this, we find $f''(x)$:

$$f''(x) = 6x - 6.$$

As such, $f''(0) = -6 < 0$, and f is concave downward at $x = 0$. Similarly, $f''(2) = 6 > 0$, and f is concave upward at $x = 2$. Noting that 2 is in the interval $[1, 15]$, we conclude that $(2, f(2)) = (2, -3)$ is an absolute minimum of f on the interval $[1, 15]$. Noting that 0 is not in the interval $[1, 15]$, we disregard this point.

Checking the endpoints, we find $(1, f(1)) = (1, -1)$ and $(15, f(15)) = (15, 2701)$. As such, we conclude that $(15, 2701)$ is an absolute maximum of f on the interval $[1, 15]$.

5. 15 points (bonus) Give an example of a function $f(x)$ for which $(x, f(x))$ is both an absolute maximum and an absolute minimum for every x in the interval $(-\infty, \infty)$.

Note that an absolute minimum of a function f on the interval $(-\infty, \infty)$ is a point $x = a$ such that $f(a) \leq f(x)$ for all $x \in (-\infty, \infty)$. Similarly, an absolute maximum of a function f on the interval $(-\infty, \infty)$ is a point $x = a$ such that $f(a) \geq f(x)$ for all $x \in (-\infty, \infty)$.

As such, for any constant function $f(x) = c$ and any point $x = a$, it follows that $f(a) = f(x)$ for all $x \in (-\infty, \infty)$.

So, if $f(x)$ is a constant function, any point $(x, f(x))$ is both an absolute maximum and an absolute minimum for every x in the interval $(-\infty, \infty)$.

Section 12.6

1. 20 points A coffee shop sells 800 cups of coffee per week at a price of \$4 each. A market survey shows that for every \$0.01 reduction in price, 3 more cups will be sold. How much should the coffee shop charge for coffee in order to maximize revenue?

First, we set up our revenue equation:

$$R(x) = (800 + 3x)(4 - 0.01x).$$

Then, we find $R'(x)$:

$$R'(x) = 4 - 0.06x.$$

Setting $R'(x) = 0$, we solve to find the root $x = \frac{200}{3}$.

Now, we find $R''(x)$:

$$R''(x) = -0.06 < 0.$$

So, we conclude that $R(\frac{200}{3}) = -0.06 < 0$ and $(\frac{200}{3}, f(\frac{200}{3}))$ is an absolute maximum of $R(x)$ on the interval $(-\infty, \infty)$. As such, we find the new, revenue-maximizing price of

$$\$ \left(4 - 0.01\left(\frac{200}{3}\right) \right) = \$\frac{10}{3} \approx \$3.33.$$

2. 20 points A box is to be made at a manufacturing company out of a square piece of cardboard 10 inches wide. Squares of equal size will be cut out of each corner, and then the ends and sides will be folded up to form a rectangular box with an open top. What size square should be cut from each corner to maximize volume?

First, we set up our volume equation:

$$V(x) = (10 - 2x)^2 x, \quad 0 \leq x \leq 5.$$

Then, we find $V'(x)$:

$$V'(x) = 12x^2 - 80x + 100.$$

Setting $V'(x) = 0$, we solve to find the roots $x = \frac{5}{3}$ and $x = 5$.

Now, we find $V''(x)$:

$$V''(x) = 24x - 80.$$

Noting that $V''(5) = 40 > 0$ and $V''(\frac{5}{3}) = -40$, and that $V(0) = 0$, we conclude that $(\frac{5}{3}, f(\frac{5}{3}))$ is an absolute maximum of V on the interval $[0, 5]$.

As such, we should cut squares with edge length $\frac{5}{3}$ inches from each corner of the cardboard square to maximize the volume of the boxes produced.